IS2218 (Finals)

AY25/26 SEM 1

github/Gavino3o

Time Value of Money

- **Future Value (FV)**: Amount an investment will grow after earning interest
- **Present Value (PV)**: Current value of a future amount
- **Discount Factor (DF)**: Present value of a $1 future payment
- **Discount Rate**: Interest rate used to compute present values of future cash flows
- **Discounted Cash Flow (DCF)**: Valuation method of calculating present value by discounting future cash flows

Perpetuities and Annuities

- **Perpetuity**: A stream of equal payments that continues forever
- **Annuity**: A stream of equal payments at regular intervals for a fixed period of time
- **Perpetuity Value**: Present value of a perpetuity, where C is the cash payment per period and r is the discount rate
- **Annuity Value**: Present value of an annuity, where C is the cash payment per period, r is the discount rate, and t is the number of periods
- **PV Annuity Factor (PVAF)**: The present value of $1 a year for each of t years
- **FV Annuity Payment (FVAP)**: Future value of a series of equal payments made at the end of each period, where C is the cash payment per period, r is the discount rate, and t is the number of periods
- **Annuity Due**: An annuity where payments are made at the beginning of each period instead of the end. Tip: Multiply the ordinary annuity value by $(1+r)$ to get the annuity due value
- **Amortization**: The process of spreading out a loan into a series of fixed payments over time
- **Amortization Schedule**: A table that details each periodic payment on an amortizing loan, showing the amount applied to principal and interest, as well as the remaining balance after each payment

Amortization Schedule

An example of an amortizing loan. If you borrow $1,000 at an interest rate of 10%, you would need to make an annual payment of $315.47 over 4 years to repay the loan with interest.

Year	Beginning-of-Year Balance	Year-End Interest Due on Balance	Year-End Payment	Amortization of Loan	End-of-Year Balance
1	$1,000.00	$100.00	$315.47	$215.47	$784.53
2	784.53	78.45	315.47	237.02	547.51
3	547.51	54.75	315.47	260.72	286.79
4	286.79	28.68	315.47	286.79	0

- **Effective Annual Rate (EAR)**: The actual annual rate that an investor earns (or pays) after accounting for compounding over a given period
- **Annual Percentage Rate (APR)**: The annual rate charged for borrowing or earned through an investment, without taking compounding into account (simple interest)
- **Monthly Rate (MR)**: The interest rate for a loan or investment calculated on a monthly basis

Inflation

- **Inflation**: The rate at which the general level of prices for goods and services is rising, leading to a decrease in purchasing power
- Current dollar cash flows must be discounted by the nominal interate rate
- **Nominal interest rate**: The stated interest rate on a loan or investment, not adjusted for inflation
- Real cash flows must be discounted by the real interest rate
- **Real interest rate**: The interest rate that has been adjusted to remove the effects of inflation, reflecting the true cost of borrowing or the true yield on an investment
- We can use inflation rate to calculate the future value of money. Just use it as the interest rate in the FV formula

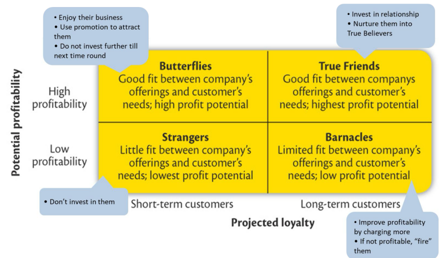
Formulas

- **Future Value,** $FV = PV(1+r)^t$
- **Present Value,** $PV = \frac{\text{Future Value after t periods}}{(1+r)^t}$
- **Discount Factor,** $DF = \frac{1}{(1+r)^t}$
- **Perpetuity Value,** $PV = \frac{C}{r}$
- **Annuity Value,** $PV = C \times \left[\frac{1-(1+r)^{-t}}{r}\right]$
- **Annuity Value,** $PV = C \times \left[\frac{1}{r} - \frac{1}{r(1+r)^t}\right]$
- **PV Annuity Factor (PVAF)** $= \left[\frac{1}{r} - \frac{1}{r(1+r)^t}\right]$
- **FV Annuity Payment (FVAP)** $= [C \times PVAF] \times (1+r)^t$
- $PV_{\text{Annuity Due}} = PV_{\text{Annuity}} \times (1+r)$
- $FV_{\text{Annuity Due}} = FV_{\text{Annuity}} \times (1+r)$
- **Loan Payment,** $C = \frac{PV \times r}{1-(1+r)^{-t}}$
- $EAR = (1+MR)^{12} - 1$
- $APR = MR \times 12$
- $1 + \text{real interest rate} = \frac{1+\text{nominal interest rate}}{1+\text{inflation rate}}$
- real interest rate $\approx$ nominal interest rate $-$ inflation rate

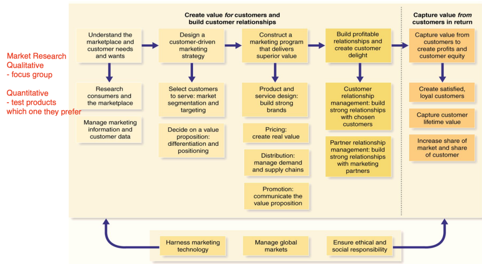
Marketing

- **Definition**: Activity, set of institutions, and processes for creating, communicating, delivering, and exchanging offerings that have value for customers, clients, partners, and society at large.
- Needs: States of felt deprivation (physical, social, individual)
- Wants: The form human needs take as they are shaped by society, culture and individual personality
- Demands: Human wants that are backed by buying power
- Marketing: Delivering Value to Customers, and in turn Capturing Value from Customers
- **Marketing Myopia**: Focusing on the product rather than the underlying customer needs
- **Production Concept:** Focus on production and distribution efficiency. Assumes consumers prefer products that are widely available and affordable. Appropriate when demand exceeds supply, or when production costs are high and improved productivity can lower prices. Impersonal, insensitive to customer needs.
- **Product Concept:** Focus on making continued product innovations. Assumes consumers favor products that offer the most quality, performance, or innovative features. Can lead to marketing myopia.
- **Selling Concept:** Focus on selling and promotion. Assumes consumers will not buy enough of the firm's products unless stimulated aggresively. Typically practiced with unsought goods (e.g. insurance, blood donations). Can lead to high risks, high costs, and unscrupulous image.
- **Marketing Concept:** Determine the needs and wants of target markets and delivering them more effectively and efficiently than competitors.
  - Crate, build, and maintain beneficial exchanges with target buyers to achieve organizational goals
  - Fcous on buyer's needs vs the seller's needs
  - Adapt and anticipate changes in consumer needs and characteristics
  - Recognise needs not just product-based
  - Stress regular consumer research and analysis
  - Resource allocation to make goods and services desired by consumers
- **Customer Driven** $\rightarrow$ Understand customers deeply about what they want $\rightarrow$ Create products that meet current needs
- **Customer Driving** $\rightarrow$ Understand customers even better than they understand themselves $\rightarrow$ Create products that meet needs now and in the future
- **Customer Relationship Management (CRM)**: Overall process of building and maintaining profitable customer relationships by delivering superior customer value and satisfaction. Not everyone is a desirable customer. Attract, keep, and grow profitable customers.

Customer Relationship Groups

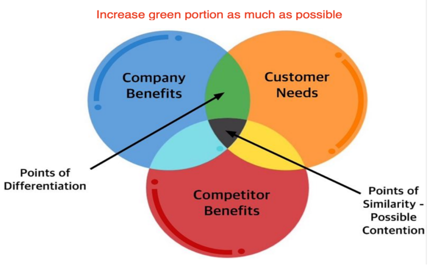


The Marketing Process



Marketing Management Process

1. **Analyse and identify market oppurtunities**
   - Marketing Research and Information Systems
   - Marketing Environment Scanning
   - Consumer and Business markets
2. **Research and select target markets**
   - Measuring and forecasting demand
   - Market segmentation, targeting, and positioning
3. **Develop marketing mix (4Ps): Product, Price, Place, Promotion**
4. **Managing the marketing effort**
   - Analysis (SWOT): Strengths, Weaknesses, Opportunities, Threats
   - Planning (Setting objectives, marketing strategy)
   - Implementation (Process of turning marketing strategy into action)
   - Control (Evaluating results, taking corrective action)



Bottom of the Pyramid (BOP)

- **Definition**: The largest, but poorest socio-economic group in the world, consisting of approximately 4 billion people who live on less than $2.50 per day.
- These markets are in the earliest stages of economic development, growth can be extremely rapid
- The competitive necessity of maintaining a low cost structure in these areas can push companies to discover creative ways to configure their products, finances, and supply chains to enhance productivity
- Hot-beds of commercial and technological experimentation

Segmentation, Targeting, Positioning (STP)

- **Segmentation**: Dividing a market into smaller groups with distinct needs, characteristics, or behaviors who might require separate products or marketing mixes. A market segment consists of consumers who respond in a similar way to a given set of marketing efforts.
- **Targeting**: Evaluating each market segment's attractiveness and selecting one or more segments to enter.
- **Positioning**: Arranging for a product to occupy a clear, distinctive, and desirable place relative to competing products in the minds of target consumers.
- **Differentiation**: Actually differentiating the market offering to create superior customer value.
- Don't have too many factors in positioning, or it will be confusing for customers
- Positioniong is perception based. Create a perception in the minds of customers about the product

Customer Lifetime Value (CLV)

1. **Operational Excellence**: Providing customers with reliable products or services at competitive prices and delivered with minimal difficulty or inconvenience. Focus on efficiency of operations and processes. (e.g. Walmart, McDonald's)
2. **Product Leadership**: Offering customers leading-edge products and services that consistently enhance the customer's use or application of the product, thereby making rivals' goods obsolete. Focus on innovation and product development. (e.g. Apple, Tesla)

3. **Customer Intimacy**: Segmenting and targeting markets precisely, then tailoring offerings to match exactly the needs of those niches. Focus on customer service and building long-term relationships. (e.g. Amazon, Nordstrom)

Aim to be best in one of the three value disciplines, and meet industry standards in the other two

Product Centric Business Model

- Great product, and producing the same at scale
- KPIs: Market share, number of new products etc
- Growth Engines: New product development, new market penetration
- Challenges: Product life cycles have shortened, competition is fierce (internet and globalization)
- Move from selling products to selling solutions (e.g entire supply chain as a service)

Customer Centric Business Model

- Who are the most valuable customers? Maximise thier long-term value, not necessarily the additinonal margin of their current purchase
- Build relationships and look into the future
- Customer Acquisition $\rightarrow$ Retention $\rightarrow$ Development

Calculating CLV

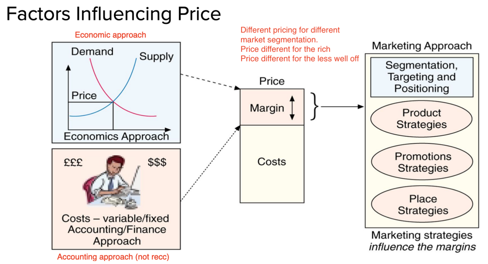
- **Definition**: The net present value of the future cash flows attributed to the customer relationship
- Use past data to estimate future customer behaviour
- Useful for customer acquisition and retention
- **Survival Rate**: Probability that a customer will remain active at the end of period t
- A **low retention rate and high discount rate**: much of the CLV comes from the first few periods
- A **high retention rate and low discount rate**: very less of the CLV is achieved in the first few periods
- **Retention rate increases over time**. Thus, it is important to calculate CLV among customers of same cohort (i.e. acquired at the same time)
- For **businesses without formal subscriptions**, other methods need to be used to evaluate retention rate, and then calculate the CLV
- Fire customers with **low or negative CLV**
- **Customer acquisition**: Spend should be **less than** the CLV of particular customer
- **Customer retention**: Spend should be **less than** the increase in CLV from retention activities.
  - $CLV_A = X$ (with retention spend, R)
  - $CLV_B = Y$ (without retention spend, $R = 0$)
  - IF $X - Y > R$, then retention spend is justified. Else, cut back on retention spend
- **Margin Expansion**: Up-sell, cross-sell, cost reduction resulting in increase of CLV to cover acquisition and retention costs

Formulas

- **CLV** $= (M - R) \times (1 + (\frac{r}{1+d}) + (\frac{r}{1+d})^2 + ...)$
- **CLV** $= (M - R) \times \frac{1+d}{1+d-r}$
  - Where:
  - M = Margin (Revenue - Cost)
  - R = Retention Spending
  - d = Discount Rate
  - r = Retention Rate
- **Survival Rate** $= r^{t-1}$
- In case of businesses where revenue is obtained after service is delivered (e.g. subscription services): **CLV** $= (M - R) \times \frac{r}{1+d-r}$
- $CLV_{\text{segment}} = CLV_{\text{avg customer}} \times$ no. of customers in segment

Pricing

- **Definition**: The amount of money charged for a product or service, or the sum of values that customers exchange for the benefits of having or using the product or service.
- Only P of the 4Ps that **generates revenue**, all others represent costs
- Inflation/Recession affects pricing strategies
- Dynamic: Prices change based on supply and demand conditions



Internal Factors Affecting Price

- **Marketing Objectives**:
  - Survival: Set prices low to cover costs and stay in business
  - Maximise current profit: Set price to maximise short-term profits, ignore long-term effects
  - Market share leadership: Set prices low to gain market share
  - Product quality leadership: Set prices high to reflect high quality
  - Competitive entry barriers: Set prices to keep out new competitors
  - Reseller support: Set prices to help resellers make profits
  - Cost recovery (non-profit & public firms): Set prices to recover costs
- **Marketing Mix Strategy**:
  - Product Design: Higher quality products can support higher prices
  - Distribution: More intermediaries increase costs and prices
  - Promotion: Heavy promotion increases costs and prices
  - Prive vs Non-Price Competition: If competitors use non-price competition, firm has more freedom to set prices. If possible, avoid price wars
  - Parrellel Importers: Prevent parallel imports by setting similar prices across countries
- **Cost Considerations**:
  - Costs set the **floor** for prices
  - Demand and competition set the **ceiling** for prices
  - Type of costs: Fixed, Variable, Total, Average, Marginal
  - Marginal cost: Variable cost of producing one more unit of a product
  - Don't set your price lower than **marginal cost**.
- **Experience Curves**:
  - As production experience increases, unit costs decrease
  - Combined effects of learning, volume, investment, and specilization lead to cost reductions (e.g . labour efficiency, product design, process engineering)
  - Exponential decay relationship between unit cost and cumulative production
- **Organisational Considerations**: Who sets the price?

External Factors Affecting Price

- **Market and Demand**:
  - Form of Competition: Pure, Monopolistic, Oligopolistic, Monopoly
  - **Price Elaticity of Demand**: Measure of sensitivity of demand to changes in price
  - Elastic Demand: Small change in price leads to large change in demand (e.g. luxury goods)
  - Inelastic Demand: Large change in price leads to small change in demand (e.g. necessities)
  - Cross-Elasticity of Demand: Measure of sensitivity of demand for product A to changes in price of product B (e.g Apple's ecosystem)
- **Consumer perception**
  - Price-Quality Relationship: Consumers associate higher prices with higher quality
  - Price used in absence of other cues: When consumers have little knowledge about a product, they rely more on price as an indicator of quality
- **Competitor's Strategies and Prices**:
  - Reference point: Consumers use competitor's prices as reference points
  - Responsiveness: Competitors may quickly match price changes
  - Product homogeneity: If products are similar, price competition is more intense/sensitive. Try to create heterogeneity to not be easily comparable
- **Other External Factors**:
  - Economy: Inflation, recession, interest rates affect consumer purchasing power
  - Distribution Channels: Intermediaries may have their own pricing strategies
  - Government: Regulations, taxes, tariffs can impact pricing decisions

Methods and Strategies

- **Price Skimming**:
-
  - Set high initial price to "skim" maximum revenues from segments willing to pay the high price, then lower price over time
  - Company makes fewer but more profitable sales
  - Effective when:
    - Segmentation on price elasticity: Some customers willing to pay high price
    - Image supportive: High price supports product's quality and image
    - Safety/Hedge: High initial price helps recover R&D costs quickly
    - Cost of smaller volume cannot be so high that it cancels the advantage of charging more
    - No immediate competition
- **Penetration Pricing**:
  - Set low initial price to penetrate market quickly and deeply, attracting a large number of buyers and market share
  - Effective when: Consumers are price sensitive, experience curve effects operative, potential competition is high
- **Cost-Oriented Pricing**:
  - Predetermined markup on cost to ensure a profit
  - Neglects demand and competitor prices
  - Markup = (Price - Cost) / Cost
  - Markup based on sales = (Price - Cost) / Price
  - **Target Return/Target Profit Pricing**: Set price to achieve a target return on investment (ROI) or target profit. Requires good demand estimates, X.
  - $P = DVC + \frac{FC}{X} + \frac{rK}{X}$. Where:
  - P = Price
  - DVC = Direct Variable Cost per unit
  - FC = Fixed Costs
  - X = Standard Volume
  - r = Desired rate of return
  - K = Capital used
- **Demand-Oriented Pricing**
- **Perceived Value Pricing**
- **Odd/Psychological Pricing**: Setting prices that have a psychological impact (e.g. $9.99 instead of $10)
- **Lost-Leader Pricing**: Setting prices below cost to attract customers to buy other products
- **Optional-Product Pricing**: Setting prices for optional or accessory products along with a main product
- **Captive-Product Pricing**: Setting prices for products that must be used along with a main product (e.g. razor and blades)
- **Bundled Pricing**: Setting a price for a set of products that is lower than the total of individual prices
- **Price Discrimination**: Time (Peak vs Off-Peak), Location (Theatre vs Home), Customer Segment (Student vs Regular)